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Introduction to: EEG

Electroencephalography (EEG) can be used to measure brain-based behavioral motivation, or how much an individual is driven to approach, or avoid, a particular stimulus.

Background

- EEG records electrical activity of the brain via electrodes placed on the scalp. It has excellent time resolution, taking hundreds to thousands of snapshots of electrical activity from multiple electrodes in a single second. This makes EEG an ideal technology to study the precise time-course of cognitive and emotional processing underlying behavior.
- To record data, an EEG headset is attached to electrodes on the scalp. The number and placement of electrodes vary based on the headset used and research goals. Generally, in addition to scalp electrodes, two electrodes are placed behind the right ear on the mastoid bone or to a clip on the ear. This process is to ensure a reference point for signal localization.
- EEG data is very prone to noise, so the best way to start is by recording clean data. The main factors that will affect the quality of data are:
    - Functional electrodes: Old electrodes must be adequately cleaned, or replaced with new electrodes when their life is up.
    - Electrode placement: Electrode placement must be consistent when recording EEG data in order to draw more representative conclusions.
    - Consistent reference: The reference site must be kept consistent. In other words, the reference cannot be switched between left and right sides.
    - Electrode impedance: Dead skin cells, oily secretions, and sweat accumulate on the scalp and can constitute a wall of electrical resistance. That's why the respondent's skin must be cleaned before the use of the EEG device.

Sampling Rate

- Once voltage has been picked up by electrodes, the continuous, analog signal has to be amplified and digitized in order to be stored on a

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computer.

- Because our brains are constantly active, there are continuous fluctuations and variations of the generated voltages. EEG systems take discrete snapshots of this continuous process, generating samples of data - similar to a picture taken by a camera. EEG systems differ in the sampling rate, or number of samples per second that they can take.
- Sampling rates are expressed in samples per second with the unit Hertz (Hz). If researchers are interested in measurements with a higher time precision, they should collect data at a higher sampling rate (i.e., > 500Hz).

Amplification

- Additionally to the digitization, the EEG signal is amplified. Think of this as the sound system for your data. Like the mono speakers on phones, poor amplification doesn't get as much signal compared to a high-end amplifier, like a movie theatre sound system, for example.
- After the signals have been digitized and amplified, they are transmitted to the recording computer through a wired connection or wirelessly.

Artefacts

- Artefacts are electrical activity that is picked up from sources in the environment rather than brain activity. Artefacts such as line noise can be removed from data using signal processing algorithms. However, if a researcher does not have resources to manually remove artefacts, it is possible to minimize their effect by ensuring a well thought out, quality lab set up, following our recommendations. It is also helpful to clearly instruct the participants on the study procedure to avoid unwanted muscle movements. For more information on artefacts, please refer to our Pocket Guide on EEG.

Because the EEG signal is sensitive to such artefacts, EEG is not recommended for certain testing scenarios. Consider your testing setup carefully when choosing EEG as a modality for your research. While we would encourage EEG for media/ad testing or other scenarios in which participants are seated in a stable position to watch video content, we would recommend against ambulatory EEG, as there is too much noise introduced in the signal in such a setup to provide valuable data.

Quantification

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There are a million ways to slice EEG data, and the decision on how to quantify and analyze your data highly depends on your research question and goals. In academia, EEG data can be analyzed on the basis of frequency (power spectral density), hemispheric dominance (frontal alpha asymmetry), communication between different areas of the brain (coherence) or by waveform in response to specific stimuli (event-related potentials), among many other ways. Commercially, frontal asymmetry (FA) and power spectral density (PSD) are two commonly used metrics.

Frontal Asymmetry

- Based on research on EEG of stable personality traits, frontal lateralization[1] reflects a person's momentary "approach-avoidance" tendencies to either engage or withdraw. Additionally, larger left-frontal band power may indicate positive emotions such as joy, while larger right-frontal band power may show negative emotional states. Recent evidence suggests that frontal lateralization can be used for testing respondents' engagement when confronted with media ads, physical products and services.
- EEG measures the differences in brain activity in the left and right frontal lobes. These scores are then normalized to put them on a readable scale.

Power Spectral Density

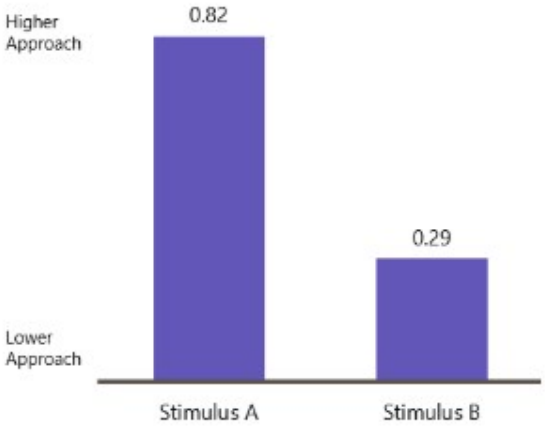
- The power of a frequency reflects the strength of the specific frequency in a signal. Higher power means that the EEG signal contains a specific frequency to a larger extent.
- Frequencies have been classified based on specific frequency bands: delta band (1-4 Hz), theta band (4-8 Hz), alpha band (8-12 Hz), beta band (13-25 Hz), and gamma band (>25 Hz). The power of these bands can be assessed to determine certain brain states, such as focused attention or deep sleep.

Data Visualizations

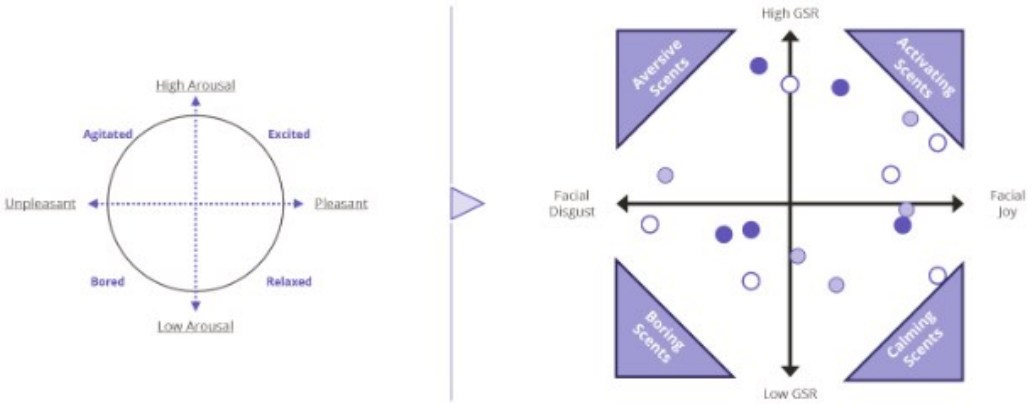
- One way to visualize EEG data is by comparing frontal asymmetry scores using a bar graph.
- In the example, Stimulus A elicits more "approach motivation" than Stimulus B: It is at the 82nd percentile in the test, while B is at the 29th percentile, suggesting that participants were more motivated by stimulus A than stimulus B. We could then combine this finding with other

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modalities to infer why that may be the case.



- Another way to visualize EEG data is by using the affective circumplex in combination with another measure. This model suggests that emotions are understood in a two-dimensional circular space, containing arousal (in the Y-axis) and valence (in the X-axis) dimensions.



- EEG data represent a measure of valence; whether the stimuli elicits more of an approach or avoid response from respondents. On the x-axis, GSR serves as a measure of arousal, or an autonomic response that indicates there is something relevant or important in the environment.
- This method of visualization allows us to view a handful of stimuli tested to compare how those stimuli can be categorized within the circumplex relative to each other. For example, a fragrance tested that falls in the upper right quadrant of the circumplex is one that elicits a high approach motivation and high arousal levels. A fragrance that falls here can be categorized as “activating.” By comparison, a fragrance that has high arousal levels and also high “avoid” motivation will fall in the upper left quadrant, and therefore be categorized as relatively “aversive” compared to other fragrances tested.

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Academic Applications

In biomedical engineering, brain-computer interfaces can be designed using frequency-based EEG metrics. By monitoring alpha activity over motor regions, researchers can detect if respondents plan to move certain limbs. These brain signals can be monitored to allow patients to steer robotic arms or legs based on their brain activity. In psychology and neuroscience, EEG is used to increase our insights on the nature of emotions, and to understand the biomarkers of mood and personality disorders. In healthcare, EEG is commonly used to assess cognitive states in neurological disorders. Those studying human factors will also use EEG to quantify cognitive workload, drowsiness and distraction.

Commercial Applications

In consumer neuroscience, EEG metrics can be monitored to investigate engagement, motivation, or drowsiness during an experience. In market research, frequency-based EEG metrics are used to analyze brand effectiveness in product packaging and design, including websites and software interfaces. EEG can also be particularly useful in advertising, trailer, and media testing. TV ads are meant to make potential shoppers buy a product/service and trailers are used to gauge interest in viewers to buy tickets to a film. Monitoring EEG activity can give unfiltered, unbiased insights into the cognitive-affective states of the audience. Similarly, online class or tutorial videos can be assessed with respect to engagement and workload metrics in order to discard material that may be hard to follow.

[1] A cognitive function being specially related to only one hemisphere of the brain.

For more in-depth explanations of EEG, please check out our blog.

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